Taylor Bosier

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Summary

Lead Data Scientist with 7+ years building production ML, Bayesian inference and optimization models that operational teams run daily, across warehousing, freight and medical devices. I also build the software around them: full-stack apps, retrieval and LLM agents, shipped with AI coding agents.

Professional Experience

Lead Data Scientist *Aug 2024 – Present*
Stryker (Remote)

- Deployed AI agents that work deduction-analyst cases (no measured results yet). Agents call a SQL data agent and statistical tools in parallel, verify each packet is complete, escalate to the associate manager over Microsoft Teams chat or call (Azure Bot Service) when stuck, and write what they learn to a vector-indexed knowledge base.
- Built an auditable Bayesian rebate-accrual forecasting platform (Python, SQL, Databricks, React/TypeScript) with Bayesian bootstrap posteriors and 95% HDIs, leakage-safe backtests against a seasonal naive baseline, and audit drill-downs. Reconciled a data-mapping discrepancy and cut report load time by 53%.
- Built search and a grounded assistant over tens of thousands of contract folders: semantic and keyword search on Databricks Vector Search, RAG chat with quoted PDF citations, and text-to-SQL behind a read-only SQL validator, so answers trace back to their source.
- Shipped 180 merged PRs in 14 months across three production web apps, directing AI coding agents through written specs, stacked PRs and two-round code review. Ported one reporting app in 42 PRs over 17 days with about 1,600 automated tests, hardened SQL parameter binding and authorization, and built the shared job scheduler the team runs its pipelines on.
- Built a causal pipeline evaluating rebate ROI with propensity score matching and inverse propensity weighting, and SKU- and family-level price optimization from isotonic demand models and dynamic programming over price tiers.
- Built contract-performance intelligence splitting year-over-year revenue change into price, volume and mix by account.
- Mentored a senior data scientist and full-stack developers; partnered with finance, operations and pricing on adoption.

Data Scientist II *Sept 2021 – Aug 2024*
Knight-Swift Transportation, Phoenix, AZ — *promoted from Data Scientist I, Sept 2023*

- Re-architected and deployed a scheduling optimization model (Gurobi) for driver planning, raising planned drivers at one terminal from 20% to 85% and improving on-time delivery and network design.
- Designed an early-warning detection model surfacing 40+ drivers a week for safety remediation.
- Conducted a study using a Bayesian Poisson-Gamma model to quantify the safety impact of lane-keeping sensors, providing critical evidence for a multi-million dollar fleet hardware decision.
- Rebuilt a driver retention model, taking F1 on the churn class from 0.20 to 0.40; the model fed the retention program that ran alongside a 12% fall in driver turnover.
- Built a frequency-severity model of fleet loss exposure that set retention and limits for the annual insurance placement.

Business Analyst *June 2019 – Sept 2021*
Chain Link Services

- Sited warehouses by k-means clustering over demand geography and forecast inventory demand with Prophet, cutting holding costs and stockouts; restructured pick paths with an apriori model and aisle-traversal heuristics, and set offsite levels with a safety-stock rule.

Technical Skills

Modelling and optimization: Python, PyMC, scikit-learn, XGBoost, LightGBM, PyTorch, PySpark, Gurobi, Pyomo
AI and software: LLM agents, RAG, vector search, Claude Code, TypeScript, Next.js, FastAPI, Rust, Docker
Data and cloud: Databricks, Azure, SQL Server, PostgreSQL, Elasticsearch, GitHub Actions, Prefect, Airflow

Open Source

rustmc: Bayesian inference library with Python models and a Rust sampler; 3,000+ PyPI downloads in six months.

Education

New Mexico State University, Las Cruces, NM — M.S. Mathematics (May 2018), B.S. Mathematics (May 2016)